Discover Civil Engineering

RESEARCH

Open Access

# Efficacy of carbon fibre reinforced polymer (CFRP) laminates in retrofitting the damaged moment resisting RC frames

Zahoor Khalid Shaikh[1*] and Vidula S. Sohoni[1]

*Correspondence:
Zahoor Khalid Shaikh
szahoor555@gmail.com
[1]Bharati Vidyapeeth (Deemed to be University) College of Engineering, Pune 411043, Maharashtra, India

**Abstract**
This study investigates the structural performance enhancement of damaged ordinary moment resisting reinforced concrete (RC) frames retrofitted with Carbon Fibre Reinforced Polymer (CFRP) laminates under seismic loading conditions. Utilizing advanced non-linear dynamic analysis in ABAQUS, the research simulates seismic-induced damage in RC frames and subsequently evaluates the effectiveness of CFRP laminates in mitigating such damage. The novelty of this work lies in its parametric evaluation of CFRP retrofit configurations, including variations in laminate thickness, length, width, and wrapping inclination angles. The findings demonstrate that a CFRP wrap with a 60° inclination outperforms the conventional 90° application in terms of displacement control, particularly at 1 mm and 2 mm thicknesses. Among various retrofitting models studied, the M2 model exhibited superior seismic displacement performance, while the M3 model showed enhanced axial load and flexural capacity. Additionally, both M2 and M3 models achieved significant improvements in shear strength compliance as per FEMA 356 guidelines. This research provides critical insights into optimizing CFRP retrofit strategies to improve the seismic resilience of existing RC frame structures.

**Keywords** Abaqus, Non-Linear analysis, Carbon fibre reinforced polymers (CFRP), Ordinary moment resisting frames

## 1 Introduction

Concrete, as a material, exhibits distinct mechanical responses under different loading conditions. Its behaviour differs significantly between static and dynamic loadings. Being inherently brittle, concrete is susceptible to failure in various circumstances. These failures may arise due to unexpected changes in loading conditions, exposure to weathering, the effects of aging, or the impact of natural disasters such as earthquakes. Some of these defects are so severe that they become irreparable or necessitate extensive and costly repair measures.

The economic impact of concrete bridge maintenance and repair is substantial, particularly in nations like the United States, Canada, Europe, and Australia. The combined annual expenditure for the upkeep and restoration of concrete bridges alone in these countries amounts to a staggering $30 billion. Abbas et al. [1], Antonopoulos et al. [2], Faella et al. [3].

Numerous research studies and approaches have been developed for the repair and rehabilitation of damaged structures, considering factors such as the material composition of the structure, the extent and severity of damage, and the feasibility of reversing the failure. Some commonly employed methods include jacketing, bracing, external plate bonding, mass reduction techniques, epoxy injection, section enlargement, and the use of fiber-reinforced polymer (FRP) wrapping or jacketing. However, traditional repair methods using concrete and steel jacketing have certain drawbacks. Their considerable weight and bulk can lead to an increase in the overall weight of rebuilt structures, potentially attracting additional loads during seismic events. Beddiar A et al. [4]. Kaveh et al. [5].

As a consequence, relying solely on conventional repair methods presents significant challenges, as they tend to be labour-intensive and expensive. Thus, there is a pressing need to explore alternative approaches that can overcome the limitations associated with the use of conventional materials in repair systems.

Over the past few decades, Fiber Reinforced Polymer (FRP) composites have experienced a surge in popularity due to their remarkable strength-to-weight ratio and their potential to mitigate durability issues commonly associated with conventional materials. The application of FRP composites as a repair technique has been widespread, aiming to restore and extend the service life of damaged infrastructure effectively. The advantages of FRP composites lie in their ease of installation, rapid deployment, ability to be manufactured under controlled conditions to ensure high quality, and their potential to act as permanent formwork in specific scenarios. As a result, prefabricated FRP repair systems have emerged as a favored and extensively adopted alternative on a global scale.

Given the high adoption rate in retrofitting applications, FRP composites have become the subject of extensive research. Numerous studies and investigations have been conducted to explore the potential and effectiveness of FRP materials in retrofitting damaged structures, and some of these studies are referenced below:

In their research work, G.M. Chen et al. [6] conducted an analytical investigation focusing on the progressive failure of Fiber Reinforced Polymer (FRP) wraps in reinforced beams. The study delved into the debonding process and subsequent rupture mechanisms, while also assessing the FRP's contribution to the beam's shear capacity. The accuracy of the analytical solution was established by comparing it with results obtained from a finite element model. One notable advantage of the analytical approach is its capability to quantitatively assess the evolution of the FRP shear contribution as shear fracture develops. This unique feature allows for a thorough examination of the shear interaction among the various components (concrete, steel, and FRP shear reinforcements) in reinforced concrete (RC) beams.

In a study conducted by N. Attari et al. [7], the effectiveness of external strengthening techniques for reinforced concrete beams using Fiber Reinforced Polymer (FRP) fabric with Glass-Carbon properties was investigated. The research aimed to address this issue by considering multiple configurations for the strengthening process. These configurations involved the use of separate, unidirectional glass and carbon fibers with U-anchorages, as well as a hybrid fabric comprising bidirectional glass and carbon

Fibers. To assess the performance of these strengthening approaches, a total of seven flexural reinforced concrete beams were subjected to testing using a 4-point bending device under various loading conditions. The objective was to conduct a comprehensive failure analysis, evaluating parameters such as strength, stiffness, ductility, and failure modes for each strengthening option considered. Additionally, the researchers developed a mathematical model capable of predicting the flexural failure of reinforced concrete components. The accuracy of this model was verified by comparing its predictions with the experimental results obtained from the testing phase. In conclusion, the study by N. Attari et al. [7] provides valuable insights into the efficacy of external strengthening methods for reinforced concrete beams using FRP fabric, specifically those with Glass-Carbon properties. The findings and the developed mathematical model contribute to a better understanding of the behaviour and failure mechanisms associated with the various strengthening configurations considered.

In the study titled "Performance of carbon fiber-reinforced polymer (CFRP)-strengthened RC frame" conducted by S. Zahoor et al. [8], structural frames were analysed using STAAD Pro software to develop a computational program for testing under various loading conditions. The frames investigated included Sound frames, Cracked frames, and CFRP-repaired wrapped frames. The study's findings yielded valuable insights into the effectiveness of CFRP in the context of strengthening and retrofitting structural elements. The results indicated that CFRP retrofitting effectively reinstates the original structural characteristics of frames, leading to enhanced lateral resistance and notable displacement capacity. Moreover, CFRP retrofitting demonstrated significant energy dissipation properties, making it a promising solution for improving the performance and resilience of existing structures. An important observation from the study was the stable cyclic behaviour exhibited by the CFRP-retrofitted frames, with minimal cumulative damage observed over repeated loading cycles. This aspect underscores the reliability and durability of the CFRP strengthening technique. In conclusion, the research highlights the potential of CFRP as a valuable tool in the domain of structural engineering, offering practical benefits for restoring and enhancing the performance of various structural elements. The study contributes to a better understanding of CFRP retrofitting's efficacy and its role in fortifying existing structures against external forces and hazards.

C D Vecchio et al. [9] conducted a study on Fiber Reinforced Polymer (FRP) applications in seismic assessment of Reinforced Concrete (RC) structural systems that were originally designed without adequate seismic details in the joint panel. The study focused on investigating the advantages of local FRP strengthening in such systems. To achieve this, the researchers devised a novel modelling approach that incorporates both the non-linear behavior of the joints and the effects of FRP strengthening using the finite element method (FEM). Various case studies were selected for the purpose of validating the proposed model. At the subassembly level, the model's predictions were compared against experimental tests conducted on full-scale beam-column joints with and without FRP strengthening. Additionally, at the structural level, a case study building that experienced joint damage during the L'Aquila earthquake and subsequently underwent retrofitting with FRP systems was analyzed. The researchers compared the predicted structural performance and damages obtained from their model with real observational data. Furthermore, the study aimed to evaluate the impact of joint FRP strengthening on the overall seismic performance of the structure by conducting numerical assessments.

Through this analysis, the researchers could quantify the benefits of FRP reinforcement on the global seismic behavior of the building.

In a study conducted by Mohamed A. El Zareef [10] [, the seismic damage behavior of multi-story reinforced concrete (RC) frame buildings constructed using lightweight concrete and beams reinforced with glass-fiber rods (GFRP) was comprehensively evaluated through numerical simulations. The primary aim of this research was to investigate the structural performance and damage response of buildings with reduced floor weight and non-metallic reinforcement under seismic loading conditions. To assess this behavior, the author employed the IDARC-2D (Inelastic Damage Analysis of Reinforced Concrete Structures) software to simulate the seismic response of five twelve-story RC frames—each differing in their use of conventional versus lightweight concrete and steel versus GFRP reinforcement—under three levels of earthquake excitations: moderate, severe, and very severe. The study systematically analysed damage indices, plastic hinge development, and displacement demands, allowing for a comparative assessment against a reference frame constructed using traditional concrete and steel reinforcement. The findings revealed that the use of lightweight concrete in floor systems effectively reduces the P–Δ effect and thereby lowers column damage indices by up to 28.5%. More notably, when GFRP rods replaced steel in beam reinforcement, beam damage indices reduced by as much as 60%, owing to the high elastic deformation capacity of GFRP, which enhanced energy dissipation without yielding. Additionally, the use of GFRP reinforcement allowed for lower post-seismic repair requirements, as bars remained within the elastic range, and only the surrounding concrete experienced cracking.

In a study conducted by Mohamed A. El Zareef and Mohamed E. El Madawy [11], the seismic performance of reinforced concrete (RC) infill frame buildings was investigated through the lens of multi-criteria optimization techniques. Recognizing the significant influence of masonry infill panels on the seismic behavior of RC frames, the study aimed to optimize both the geometric and mechanical properties of the infill panels to enhance the overall seismic response of a 12-story RC frame structure. The authors adopted a comprehensive approach by integrating the Parameter Space Investigation (PSI) method with inelastic damage analysis using the IDARC-2D software. A total of 500 design variable combinations were generated using LPτ-sequences, focusing on four key parameters of the masonry infill panels: maximum compressive strength, strain at peak compressive strength, shear strength, and wall thickness. These combinations were analysed under nonlinear dynamic conditions using ground motion data from the 1940 El Centro earthquake. Through this analytical framework, the study evaluated multiple seismic performance criteria, including story displacement, drift, shear, base shear, and overall damage index. The optimization results revealed that certain configurations of masonry properties significantly reduce story drift and top-floor displacements, and improve shear resistance. For instance, optimal infill panel properties resulted in a 42.8% reduction in average story drift and a 45.46% reduction in maximum displacement when compared to the least favorable configuration. The findings also emphasized the critical role of wall thickness, with panels of 200 mm or more contributing substantially to in-plane stiffness and energy dissipation.

Based on the results obtained from the pushover analysis, the extent of structural damages was identified. Subsequently, the identified damages were addressed through the application of CFRP wraps for retrofitting purposes. Notably, the CFRP wraps were strategically installed at a specific angle of 60 degrees. This choice was based on its superior efficiency compared to wraps installed at a 90-degree angle, leading to improved structural strengthening and enhanced performance of the frame.

### 1.1 Numerical analysis

For the numerical simulation, three distinct models were created and analyzed using the finite element (FE) software ABAQUS v.4.17. The analysis involved a parametric study where the thickness of the Carbon fiber-reinforced polymer (CFRP) sheet, the angle of its installation, and the length of its application were systematically varied and investigated. To address the significant weakness in the concrete's tensile zone, and following the guidelines outlined in the ACI 440.2R-08 recommendation, the models were reinforced with CFRP sheets positioned at the bottom of the beam, extending along its entire length. The specific characteristics and properties considered for these FRP sheets are outlined in detail in Table 1.

## 2 Methodology

For performing Non-linear Pushover Analysis in ABAQUS, the nonlinear behaviour of the structure was defined with the "concrete damage plasticity "clause. The compression and tensile diagram were defined in a tabular form, and then the first step (static general) was created where the gravity loads were applied. In the second step (static general) where body forces (NN/m3) proportional to the mass were applied. The body forces were applied in the X Direction, in order to simulate the seismic action. In order to create an increasing lateral loading, the linear distribution function was used. After

**Table 1** Properties of the modelled structures

| **Structural/Geometric properties of the model** | |
|---|---|
| FRP | Carbon Fibre Reinforced Polymer (CFRP) |
| Thicknesses Used | 1 mm/2 mm |
| Shape of Installation | U-Shaped |
| Angle of installation | $60^{o}/90^{o}$ |
| Width of Wrap | 150 mm |
| Spacing between Wraps | 150 mm |
| Length of Wrap Beyond Damaged Section | 600 mm on Either Sides |
| Compressive Strength of Concrete | 40 MPa. |
| Column Sizes | 230 × 530 (mm) |
| Beam Sizes | 230 × 450 (mm) |
| Length of Each Bay | 3000 (mm) |
| Height of Each Bay | 3000 (mm) |
| Clear Cover of Concrete | 25 mm |
| Rebars in Beams & Columns | 16 mm |
| Stirrups | 8 mm |
| Mechanical Properties of the Model | |
| Strain in concrete (for max strength) | 0.00185932 |
| Modulus of Elasticity of Concrete | 18.01148 MPa. |
| Modulus of Elasticity of Rebar Steel | 200 GPa. |
| Damage for Traction Separation Laws (Max PS Damage) | 22 GPa. |
| Mechanical Properties of FRP | |
| Tensile Strength of FRP | 2000–3600 MPa. |
| Modulus of Elasticity | 125–600 MPa. |
| Strain | 1.2 |

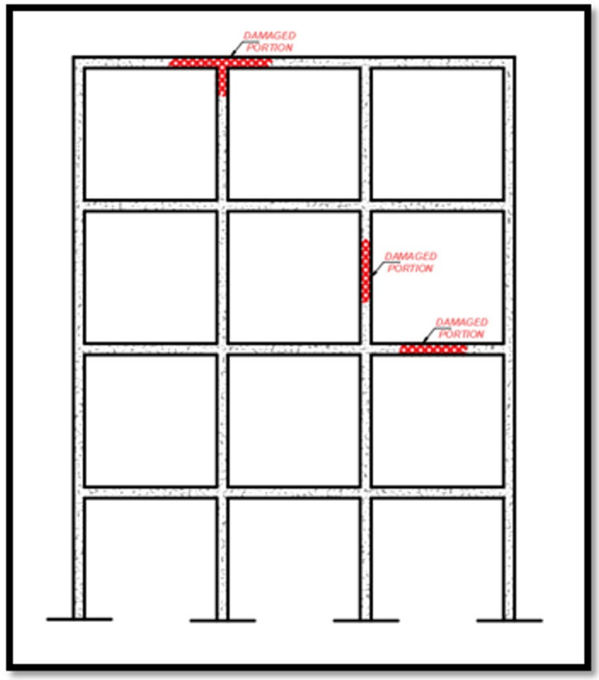


**Fig. 1** Illustration of Damaged Section-1

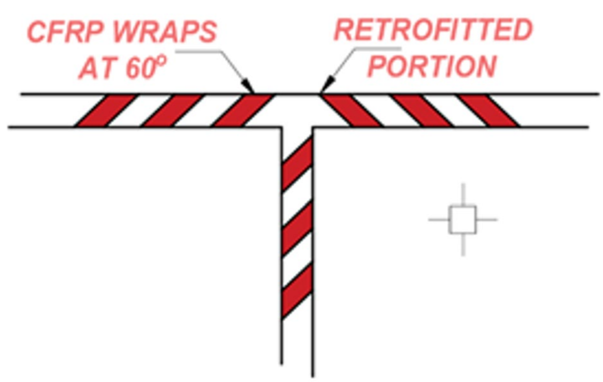


**Fig. 2** Illustration of Damaged Section-2

performing the analysis Stresa, Von-Mises Stresses were checked and crack/failure initiation at the plastic hinges were checked. The location of plastic hinges was wrapped by CFRP sheets.

CFRP Sheets were modelled using 'Rebar-Layer'. Spacing of rebar in the plane of the shell element was defined. The value is given as an Angular measure as the GEOMETRY = ANGULAR parameter is specified; in which case the value should be given in terms of angular spacing in degrees. The part of CFRP Wrapped concrete was defined as secured and remaining was defined as unsecured. The spacing was specified in the uncured geometry if GEOMETRY = LIFT EQUATION. The local 1-direction was defined by using the ORIENTATION option and setting the ORIENTATION parameter equal to the orientation name. The isoperimetric direction was set to the default value. The extension ratio, e, for rebar was defined with GEOMETRY = LIFT EQUATION. Figure 1 shows the illustration of Damaged Section-1, while Fig. 2 presents the illustration

**Table 2** Details of the models simulated

| Model designation | Details of the model |
|---|---|
| M1 | Intact Model with Earthquake Loadings and Pushover Analysis |
| M2 | Damaged Model with CFRP Wrapped 1 mm thickness at 60° and 90°Angle |
| M3 | Damaged Model with CFRP Wrapped 2 mm thickness at 60° and 90°Angle |

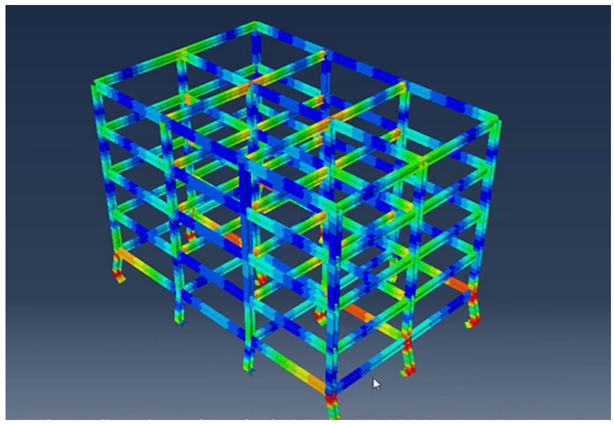

**Fig. 3** Extruded View of M1 Model in ABAQUS

**Table 3** Displacement values of 1 mm Thick CFRP wrapped models at 60° and 90°

| Max displacement | Intact model (mm) | 1 mm thickness at 60° | 1 mm thickness at 90° |
|---|---|---|---|
| 0 | 28.617 | 28.048 | 31.209 |
| 3 m | 34.016 | 33.373 | 36.973 |
| 6 m | 40.854 | 41.748 | 45.109 |
| 9 m | 49.375 | 52.498 | 55.849 |

of Damaged Section-2, highlighting the typical failure patterns observed in ordinary moment resisting frames under seismic loading.

The details of the models simulated are specified as under in Table 2:

## 3 Investigation of the results

1. Displacement values

Displacement is one of the primary results obtained from an FEA simulation. Displacement denotes structural deformation under loading. It is a vector quantity that describes the change in position of each node in the finite element model. The displacement value indicates the magnitude and direction of the deformation at each node (Fig. 3).

In the present study these displacements values allowed to understand how the structure responds to the applied loads and identify critical areas with large deformations or potential failure points. In summary, displacement values obtained provided valuable insights into the structural behaviour and deformation characteristics of a models M1, M2 and M3 under applied loads (Tables 3 and 4) (Figs. 4 and 5).

where,

M1 is intact Model with Earthquake Loadings and Pushover Analysis.

M2 is damaged Model with CFRP Wrapped 1 mm thickness at 60° and 90 ° Angle.

M3 is damaged Model with CFRP Wrapped 2 mm thickness at 60 ° and 90 ° Angle.

**Table 4** Displacement values of 2 mm Thick CFRP wrapped models at 60° and 90°

| Max displacement | Intact model | 2 mm thickness at 60° | 2 mm thickness at 90° |
|---|---|---|---|
| 0 | 18.687 | 27.547 | 34.534 |
| 3 m | 22.212 | 32.882 | 34.448 |
| 6 m | 26.678 | 39.260 | 33.749 |
| 9 m | 32.242 | 48.005 | 34.201 |

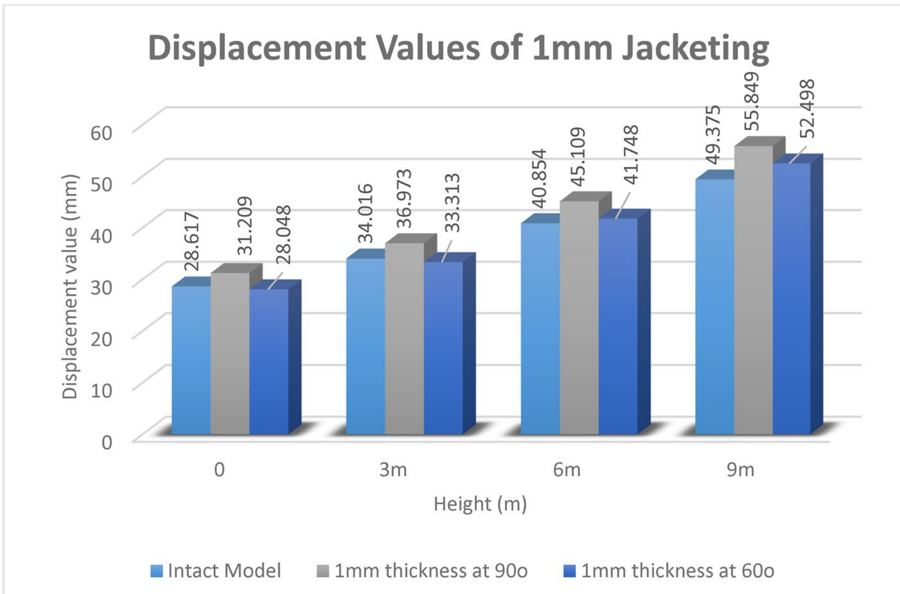


**Fig. 4** Graph 1: Graphical Representation of Displacement Values of 1 mm thick CFRP wrapped Models at 60°and 90°

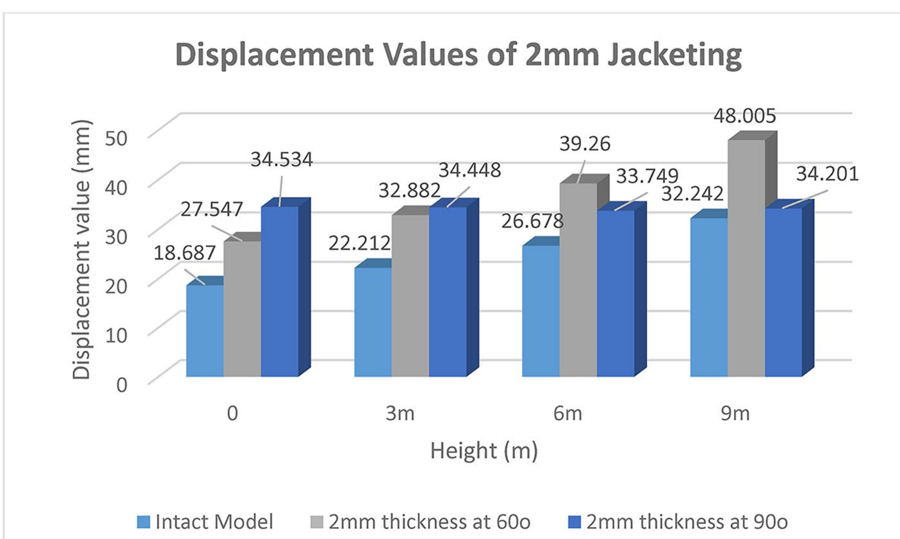


**Fig. 5** Graph 2: Graphical Representation of Displacement Values of 2 mm thick CFRP wrapped Models at 60° and 90°

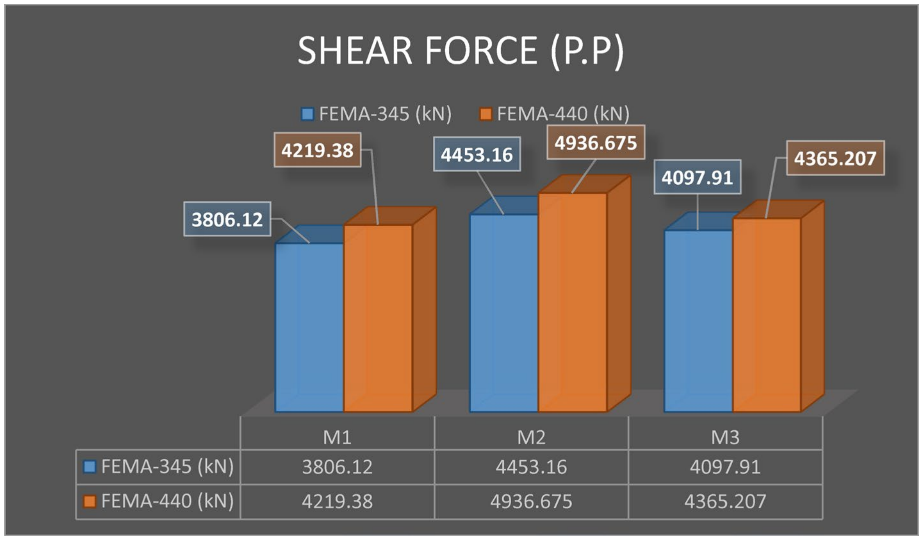


**Fig. 6** Graph 3: Graphical Representation of Performance Point (Shear Force)

**Table 5** Performance point (Shear Force)

| Performance Point (SF) | FEMA-345 (kN) | FEMA-440 (kN) |
|---|---|---|
| **M1** | 3806.12 | 4219.38 |
| **M2** | 4453.16 | 4936.675 |
| **M3** | 4097.91 | 4365.207 |

Considering the favourable outcomes obtained from the application of a 60-degree inclination wrapping technique on Carbon fiber-reinforced polymer (CFRP) specimens with thicknesses of 1 mm and 2 mm, specifically in relation to Displacement Values, it is therefore deemed pertinent to exclusively reference the results associated with the 60$^{o}$ inclination wrapping method for the subsequent comparative assessment of parameters. These parameters encompass Performance Point (Shear Force), Performance Point (Max Displacement), Axial Load Capacity of Columns, and Flexural Strength of Beams within the context of the research study.

2. Performance point (Shear force)

Performance Point (Shear Force), is a crucial output parameter in finite element analysis (FEA) simulations involving structures subjected to various loading conditions. Shear Force is a vector quantity that describes the internal forces within a structure perpendicular to its longitudinal axis (Fig. 6).

When in the present study the analysis was done in Abaqus, the software calculated the Shear Force at each node in the finite element model. The magnitude and direction of the Shear Force represent the intensity and orientation of the internal forces that cause shear deformation in the structure. Shear forces are particularly significant in beams, columns, and other structural elements, where they can lead to bending and failure if they exceed the material's strength limits. In summary, Shear Force, as a Performance Point is a fundamental output parameter used to analyse and evaluate the structural behaviour of components and systems (Table 5).

where,

M1 is intact Model with Earthquake Loadings and Pushover Analysis.

M2 is damaged Model with CFRP Wrapped 1 mm thickness at 60º and 90 º Angle.

M3 is damaged Model with CFRP Wrapped 2 mm thickness at 60 º and 90 º Angle.

3. Performance point (Max displacement)

Performance Point (Max Displacement), is a critical output parameter in finite element analysis (FEA) simulations. Max Displacement represents the maximum magnitude of displacement experienced by a structure under applied loads.

When performing the analysis of present study in Abaqus, the software calculated the displacement values at each node in the finite element model. The maximum displacement suggested the highest value among all these calculated displacements. It indicates the most significant deformation that occurs within the structure.

The Max Displacement is a crucial result in FEA as it provides valuable insights into the structural behaviour and deformation characteristics of the model under the given loading conditions. This information is used to assess whether the structure can withstand the applied loads without exceeding its allowable deformation limits. Large displacements may lead to buckling, stress concentrations, or potential failure, so identifying and addressing regions with excessive displacement is essential for ensuring the structural integrity and safety of the design (Table 6).

In summary, Max Displacement, as a Performance is a crucial output parameter used to assess the extent of deformation and potential failure points in a finite element model (Fig. 7).

where,

M1 is intact Model with Earthquake Loadings and Pushover Analysis.

M2 is damaged Model with CFRP Wrapped 1 mm thickness at 60º and 90 º Angle.

M3 is damaged Model with CFRP Wrapped 2 mm thickness at 60 º and 90 º Angle.

4. Axial load capacity of columns

The axial load capacity of columns is a vital aspect in structural engineering and is particularly relevant in the design and analysis of columns subjected to axial loads. Abaqus software, a finite element analysis (FEA) tool, can be used to determine the axial load capacity of columns accurately.

In structural engineering, columns are vertical structural elements that primarily carry axial loads, i.e., loads applied along their longitudinal axis. The axial load capacity of a column refers to the maximum amount of compressive load it can sustain before experiencing failure, such as buckling or excessive deformation (Figs. 8 and 9).

Once the finite element model is set up, Abaqus can perform a static or nonlinear analysis to calculate the column's response to the applied axial loads. By gradually increasing the applied axial load, you can determine the critical load at which the column reaches its limit state, leading to buckling or failure. Post-processing tools in Abaqus allow you

**Table 6** Performance point (Max Displacement)

| Performance point (SF) | FEMA-345 (kN) | FEMA-440 (kN) |
|---|---|---|
| **M1** | 306.74 | 449.28 |
| **M2** | 286.78 | 317.36 |
| **M3** | 314.86 | 361.854 |

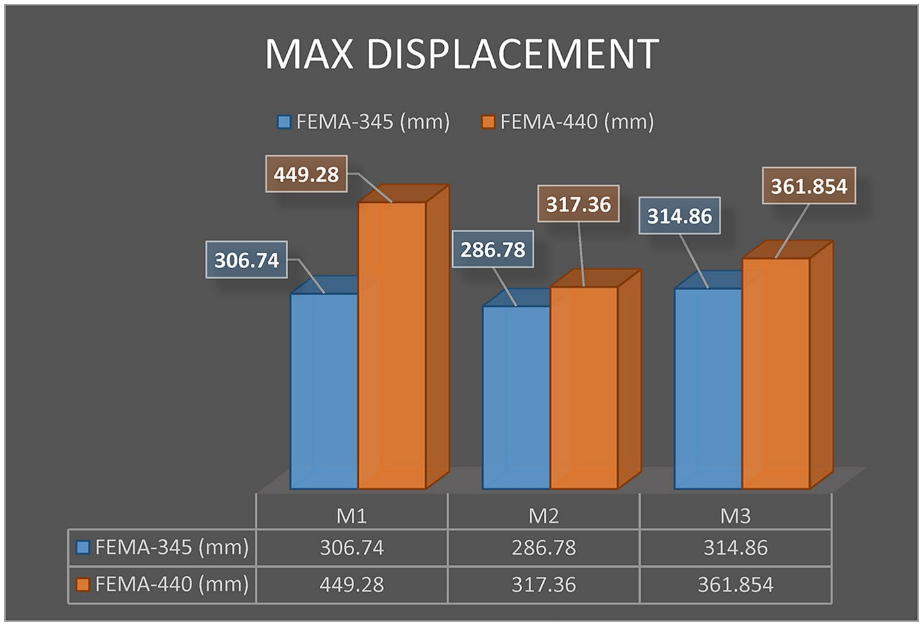


**Fig. 7** Graph 4: Graphical Representation of Performance Point (Max Displacement)

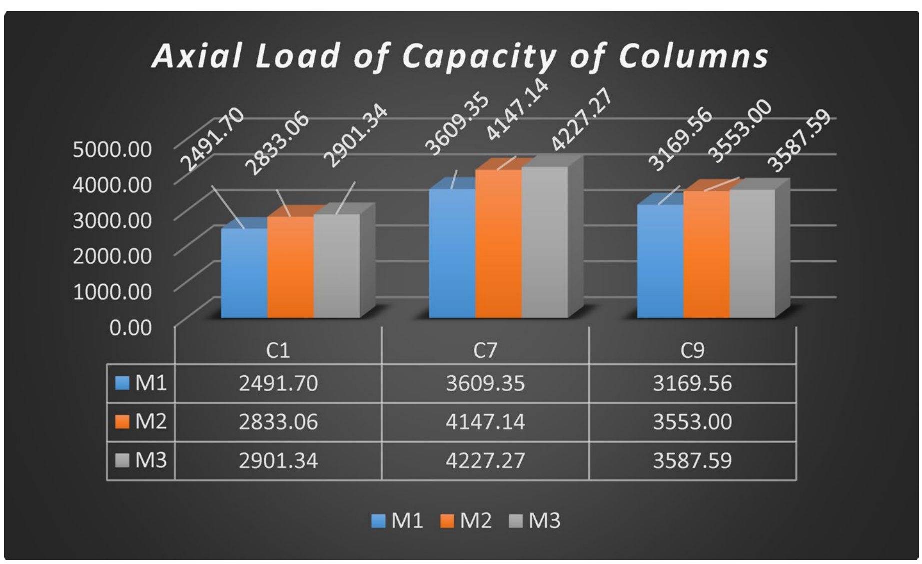


**Fig. 8** Graph 5: Graphical Representation of Axial Load Capacity of Columns

to visualize and analyze the results, such as the deflection, stress distribution, and mode of failure (Table 7).

5. Flexural strength of beams

The flexural strength of beams is a crucial mechanical property that characterizes a beam's ability to resist bending before experiencing failure. In Abaqus software, a widely used finite element analysis (FEA) tool, engineers and researchers can determine the flexural strength of beams through simulation and analysis. Flexural strength is particularly important in the design and analysis of structural components, such as beams and girders, which are subjected to bending loads. The ability to withstand these loads

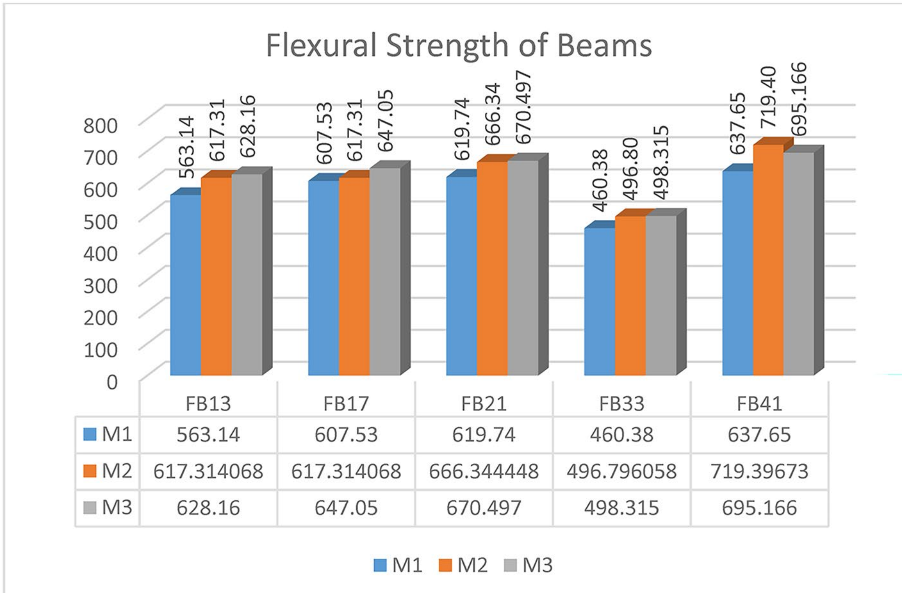


**Fig. 9** Graph 6: Graphical Representation of Flexural Strength of Beams

**Table 7** Axial load capacity of columns

| Critical columns | M1 (kN) | M2 (kN) | Capacity Increase (%) | M3 (kN) | Capacity Increase (%) |
|---|---|---|---|---|---|
| C1 | 2491.7 | 2833.063 | 13.7 | 2901.335 | 16.44 |
| C7 | 3609.35 | 4147.143 | 14.9 | 4222.218 | 16.986 |
| C9 | 3169.56 | 3553.077 | 12.13 | 3587.593 | 13.22 |

**Table 8** Flexural strength of beams

| Critical beams | M1 (kN.m) | M2 (kN.m) | Capacity increase (%) | M3 (kN.m) | Capacity increase (%) |
|---|---|---|---|---|---|
| FB13 | 563.14 | 617.3141 | 9.62 | 628.1264 | 11.54 |
| FB17 | 607.53 | 644.7716 | 6.13 | 647.5055 | 6.58 |
| FB21 | 619.74 | 666.3444 | 7.52 | 670.4967 | 8.19 |
| FB33 | 460.38 | 496.7961 | 7.91 | 498.3153 | 8.24 |
| FB41 | 637.65 | 719.3967 | 12.82 | 695.166 | 9.02 |

without excessive deformation or failure is essential for the safety and reliability of engineering structures.

To determine the flexural strength of a beam using Abaqus, you create a finite element model that accurately represents the beam's geometry and material properties. The model should consider all relevant boundary conditions and loading scenarios that simulate real-world conditions, including the application of bending moments. During the analysis, Abaqus calculates the deflection, stress distribution to evaluate the beam's behaviour.

Furthermore, Abaqus allows users to perform failure analysis to predict the ultimate strength and failure mode of the beam. This information is valuable for optimizing the beam's dimensions, material selection, or reinforcement strategy to enhance its flexural strength (Table 8).

## 4 Conclusions

In conclusion, this research study aimed to investigate the use of CFRP wraps on inclinations of 60 and 90 degrees to enhance the performance of structural elements. The decision to choose these inclinations was based on previous studies that suggested they exhibited the best results. The research employed Pushover analysis to draw the following conclusions:

- The results indicated that when using a 1 mm thickness and 2 mm thickness of CFRP wrapping, the inclination of 60 degrees showed better displacement performance than the 90-degree inclination. This suggests that the effectiveness of the CFRP wrap depends on both inclination angle and material thickness.
- Regarding shear force results, the M2 and M3 models outperformed others, particularly showing better results with respect to FEMA 340 guidelines. This implies that these models are more effective in enhancing shear force performance, which is crucial for the structural integrity of the elements.
- The M2 model demonstrated the best displacement (performance point) results, with the least amount of displacement observed. This indicates that the M2 model is the most effective in limiting deformations and maintaining the stability of the structural elements.
- Finally, in terms of enhancing axial load carrying capacity and flexural strength, the M3 model performed better than the M2 model. This suggests that the M3 model offers higher resistance to axial loads and bending, making it more suitable for enhancing the overall structural performance.

In summary, this research provides insights into the use of CFRP wraps at different inclinations and thicknesses for enhancing the performance of structural elements. The results highlight the importance of selecting the appropriate inclination and thickness to optimize the structural behaviour and ensure safe and resilient structures in real-world applications. Further research and testing may be needed to explore the effects of other parameters and to validate the findings in various structural configurations.

### 4.1 Limitations

While this study offers valuable insights into the structural performance enhancement of RC frames retrofitted with CFRP laminates at different inclinations, several limitations should be acknowledged:

- Long-term durability:

The study focuses on short-term structural performance and does not assess the long-term durability of CFRP wraps under sustained loading or aging. The effects of creep, fatigue, and material degradation over time were beyond the scope of this research.

- Environmental Exposure:

The analysis does not account for environmental factors such as temperature fluctuations, moisture ingress, freeze-thaw cycles, or chemical exposure. These factors can significantly influence the bond behavior and effectiveness of CFRP materials in real-world applications.

- Cyclic and Fatigue Loading:

Although the models were evaluated using nonlinear static pushover analysis, the effects of cyclic loading representative of repeated seismic or operational events were not explicitly studied. Further investigations incorporating full-scale cyclic testing are necessary to understand the degradation behavior of inclined CFRP wraps over multiple loading cycles.

- Material and Geometric Variability:

The current models assume ideal material properties and consistent geometry. Variations due to construction tolerances, defects, or site conditions could affect the actual performance of the retrofitting system.

- Field Implementation Constraints:

While the study demonstrates the technical viability of 60° CFRP inclination, practical challenges such as labor training, access in confined locations, or curvature of structural elements were not addressed.

Future research should consider these limitations through long-term experimental programs, environmental simulation studies, and full-scale cyclic performance evaluations to validate and extend the findings presented herein.

## Supplementary Information

The online version contains supplementary material available at https://doi.org/10.1007/s44290-025-00326-5.

Supplementary Material 1

**Acknowledgements**

The first author acknowledged the support of KRISHNA CONCHEM PRODUCTS PVT LTD, Mumbai and Shri Tulsi Buildcon, Aurangabad for their constant support during the research work.

**Author contributions**

Shaikh Zahoor Khalid (first author) proposed the formulation in this paper and wrote the manuscript and draw figures. Dr. Vidula S. Sohoni (second author) edited the manuscript.

**Funding**

For the preparation of this manuscript and conducting this research, no fund was received.

**Data availability**

The datasets generated during and/or analysed during the current study are available from the corresponding author on reasonable request.

## Declarations

**Ethics approval and consent to participate**

This study did not involve human participants or animals. Hence, ethical approval was not required. Not applicable, as no human participants were involved in the study.

**Consent for publication**

All authors have read and approved the final version of the manuscript and consent to its submission for publication.

**Competing interests**

The authors declare no competing interests.

Received: 14 April 2025 / Accepted: 12 September 2025

Published online: 19 September 2025

**References**

1. Abbas AA, Mohsin SMS, Cotsovos DM. Seismic response of steel fibre reinforced concrete beam–column joints. Eng Struct. 2014;59:261–83. https://doi.org/10.1016/j.engstruct.2013.10.046.
2. Antonopoulos CP, Triantafillou TC, Asce M. (2002). Analysis of FRP-strengthened RC beam–column joints. Journal of Composites for Construction. https://doi.org/10.1061/(ASCE)1090-0268(2002)6:1(41).

3. Faella C, Martinelli E, Nigro E. Formulation and validation of a theoretical model for intermediate de-bonding in FRP-strengthened RC beams. Compos Part B Eng. 2008;39:645–55. https://doi.org/10.1016/j.compositesb.2007.06.002.
4. Beddiar A, Zitoune Rédouane, Collombet F, Grunevald Y-H, Abadlia M, Bourahla N. Compressive behaviour of concrete elements confined with GFRP-prefabricated bonded shells. Eur J Environ Civil Eng. 2014. https://doi.org/10.1080/19648189.2014.929538.
5. Kaveh A, Bakhshpoori T, Hamze-Ziabari SM. Computers Concrete. 2018;22(2):197–207. https://doi.org/10.12989/cac.2018.22.2.197. GMDHbased prediction of shear strength of FRP-RC beams with and without stirrups.
6. Chen G, Li S, Fernando D, Liu P, Chen J. Full-range FRP failure behaviour in RC beams shear-strengthened with FRP wraps. Int J Solids Struct. 2017;125:1–21. https://doi.org/10.1016/j.ijsolstr.2017.07.019.
7. Attari N, Amziane S, Chemrouk M. Flexural strengthening of concrete beams using CFRP, GFRP and hybrid FRP sheets. Constr Build Mater. 2012;37:746–57. https://doi.org/10.1016/j.conbuildmat.2012.07.052.
8. Shaikh ZK, Sohoni VS. (2023, March 6). Performance of carbon fiber-reinforced polymer (CFRP)-strengthened RC frame. Asian Journal of Civil Engineering, 24(6), 1831–1840. https://doi.org/10.1007/s42107-023-00609-1
9. Del Vecchio C, Di Ludovico M, Prota A, Manfredi G. Modelling beam-column joints and FRP strengthening in the seismic performance assessment of RC existing frames. Compos Struct. 2016;142:107–16. https://doi.org/10.1016/j.compstruct.2016.01.077.
10. Zareef MaE. Seismic damage assessment of multi-story lightweight concrete frame buildings reinforced with glass-fiber rods. Bull Earthq Eng. 2016;15(4):1451–70. https://doi.org/10.1007/s10518-016-0027-0.
11. Zareef MaE, Madawy MEE. Optimization of infill panel for seismic response of multi-story RC frame buildings utilizing multi criteria optimization technique. Bull Earthq Eng. 2018;16(10):4951–70. https://doi.org/10.1007/s10518-018-0363-3.

## Publisher's note